

MI-09: Partial Result

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Repository status: Partially resolved. The complete argument received an independent Codex-agent PASS review for the stated partial result only. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope.](#)

1 MI-09, dimension-two subproblem: an exact sharp universal norm constant

Write $\mathcal{S}(T) = (|T| + |T^*|)/2$. We determine the sharp constant uniform over all unitarily invariant norms in dimension two. This resolves the dimension-two question in [1, Problem 1.11], including its operator-norm case, but does not determine the separate optimal constants for every Schatten exponent and every matrix dimension in the broader MI-09 program.

Theorem 1.1. *For every $m \geq 2$, every $T_1, \dots, T_m \in \mathbb{M}_2(\mathbb{C})$, and every unitarily invariant norm $\|\cdot\|$,*

$$\|\mathcal{S}(\sum_j T_j)\| \leq \sqrt{6\sqrt{3} - 9} \|\sum_j \mathcal{S}(T_j)\|.$$

The constant $\sqrt{6\sqrt{3} - 9} = 1.1799596795\dots$ is optimal when the inequality is required for all such norms. It is already attained in the operator norm by two real rank-one matrices. In fact, the sharp numerical-radius inequality is

$$\|\mathcal{S}(T)\|_\infty \leq \sqrt{6\sqrt{3} - 9} w(T), \quad T \in \mathbb{M}_2(\mathbb{C}), \quad (1)$$

where $w(T) = \max_{\|v\|=1} |v^* T v|$.

1.1 Reduction to a numerical-radius inequality

For every matrix Z and unit vector v , the polar decomposition and Cauchy–Schwarz give

$$|v^* Z v| \leq \sqrt{(v^* |Z| v)(v^* |Z^*| v)} \leq v^* \mathcal{S}(Z) v.$$

Consequently, with $T = \sum_j T_j$ and $D = \sum_j \mathcal{S}(T_j)$,

$$w(T) \leq \|D\|_\infty.$$

Thus (1) implies the claimed operator-norm inequality. Also

$$\mathrm{tr} \mathcal{S}(T) = \|T\|_1 \leq \sum_j \|T_j\|_1 = \mathrm{tr} D.$$

The largest-eigenvalue inequality and this trace inequality imply weak majorization of the two eigenvalues of $\mathcal{S}(T)$ by those of $\sqrt{6\sqrt{3} - 9} D$. Hence the inequality holds for every unitarily invariant norm.

1.2 A normal form adapted to the symmetric modulus

For a nonzero 2×2 matrix, set $q = \|T\|_1/2$. The elementary two-dimensional square-root identity yields

$$\mathcal{S}(T) = \frac{T^*T + TT^* + 2|\det T|I}{2\|T\|_1}. \quad (2)$$

Indeed, for a positive 2×2 matrix P , its positive square root equals $(P + \sqrt{\det P} I)/\sqrt{\operatorname{tr} P + 2\sqrt{\det P}}$; the formula extends to singular matrices by continuity.

Multiplication by a scalar of modulus one preserves both sides of (1), as does unitary similarity. Make $\operatorname{tr} T = 2h$ real and positive, and diagonalize $\mathcal{S}(T)$ in decreasing order. The off-diagonal entry in the numerator of (2) is $2h(t_{12} + \bar{t}_{21})$, so $t_{21} = -\bar{t}_{12}$. A diagonal unitary similarity now gives

$$T = \begin{pmatrix} h+x+ik & b \\ -b & h-x-ik \end{pmatrix}, \quad h, x, b \geq 0, \quad k \in \mathbb{R}. \quad (3)$$

The ordering of the symmetric-modulus eigenvalues gives $x \geq 0$. Cases with zero trace or a repeated symmetric-modulus eigenvalue follow by continuity from these generic cases.

Let $u = h^2 + x^2 + k^2 + b^2$. Direct calculation gives

$$q^2 = \frac{u + |\det T|}{2}, \quad \mathcal{S}(T) = \operatorname{diag} \left(q + \frac{hx}{q}, q - \frac{hx}{q} \right).$$

Moreover,

$$(q^2 - x^2)(q^2 - h^2 - b^2) = q^2 k^2.$$

We have $h \leq q$ by $|\operatorname{tr} T| \leq \|T\|_1$, and $x \leq q$ follows from the displayed formula for q^2 (or by evaluating its quadratic equation at x^2). By homogeneity take $q = 1$. By continuity we may assume $0 < h, x < 1$. Then

$$b^2 = 1 - h^2 - \frac{k^2}{1 - x^2}, \quad \|\mathcal{S}(T)\|_\infty = 1 + hx. \quad (4)$$

For the normal form (3), the numerical range is the filled ellipse

$$\{h + xs + i(ks + bt) : s^2 + t^2 \leq 1\}.$$

This follows directly by writing the expectation of the real and imaginary Hermitian parts on a unit vector in $\mathbb{C}^2$.

1.3 A sharp lower bound for the numerical radius

Put

$$H = h^2, \quad X = x^2, \quad P = 1 - X, \quad Q = 1 - H, \quad g = 1 - H - X, \quad z = k^2.$$

First suppose $g \geq hx$. We claim

$$w(T)^2 \geq \frac{(1-H)(1-X)}{1-H-X} = 1 + \frac{HX}{g}. \quad (5)$$

The generic case has $g > 0$. To prove the claim, set

$$M = \begin{pmatrix} X+z & kb \\ kb & b^2 \end{pmatrix}, \quad a = \begin{pmatrix} hx \\ 0 \end{pmatrix}.$$

The ellipse formula gives the elementary quadratic maximization identity

$$w(T)^2 = H + \inf_{\lambda > \lambda_{\max}(M)} \{ \lambda + a^T (\lambda I - M)^{-1} a \}. \quad (6)$$

Here the infimum may be attained only as a boundary limit. For completeness, completion of the square gives the upper bound in (6) for every $\lambda > \lambda_{\max}(M)$. Equality holds when $v = (\lambda I - M)^{-1}a$ has norm one. Its norm decreases continuously to zero as $\lambda \rightarrow \infty$. If it does not reach one before the lower endpoint, its endpoint value has norm at most one and is orthogonal to a top eigenvector; adding a suitable multiple

of that eigenvector gives a unit vector achieving equality in the endpoint limit. This proves (6), including that boundary case.

Equation (4) gives

$$\det(QI - M) = -\frac{HXz}{P} \leq 0.$$

Since $Q > X$ here, it follows that $\lambda_{\max}(M) \geq Q$. Write $\lambda = Q + d$ with $d > 0$, and put

$$\Delta = \det(\lambda I - M) = d(g + d) + \frac{zX(d - H)}{P} > 0.$$

The expression inside the infimum, including H , becomes

$$f = 1 + d + \frac{HX(d + z/P)}{\Delta}.$$

The following exact polynomial identity is the key estimate:

$$\begin{aligned} g\Delta \left(f - 1 - \frac{HX}{g} \right) &= d^2(g^2 - HX + gd) \\ &\quad + \frac{zX}{P}(gd^2 - HQd + HQP). \end{aligned}$$

The first term is nonnegative because $g \geq hx$. Also $HQ \leq gP$, since $gP - HQ = g^2 - HX \geq 0$. If $0 \leq d \leq P$, then $gd^2 - HQd + HQP = gd^2 + HQ(P - d) \geq 0$. If $d \geq P$, then $gd^2 - HQd + HQP = d(gd - HQ) + HQP \geq 0$. Thus $f \geq 1 + HX/g$ for every admissible λ . Taking the infimum proves (5).

Let $p = hx$. The condition $g \geq p$ implies $0 \leq p \leq 1/3$. Since $H + X \geq 2p$, equation (5) yields

$$\frac{\|\mathcal{S}(T)\|_\infty^2}{w(T)^2} \leq \frac{(1+p)^2 g}{g+p^2} \leq \frac{(1+p)^2(1-2p)}{(1-p)^2}.$$

The last rational function on $[0, 1/3]$ has its maximum at $p = 2 - \sqrt{3}$, where its value is $6\sqrt{3} - 9$. This follows by differentiating: the only critical point in that interval solves $p^2 - 4p + 1 = 0$.

It remains to consider $g \leq hx$. Now $w(T) \geq h + x$, using the real part of T . If $p = hx \leq 1/3$, then

$$(h+x)^2 = H + X + 2p \geq 1 + p, \quad \frac{(1+p)^2}{(h+x)^2} \leq 1 + p \leq \frac{4}{3}.$$

If $p \geq 1/3$, then

$$\frac{(1+p)^2}{(h+x)^2} \leq \frac{(1+p)^2}{4p} \leq \frac{4}{3} \quad (1/3 \leq p \leq 1).$$

Since $4/3 < 6\sqrt{3} - 9$, this proves (1) in all cases.

1.4 Two explicit rank-one summands attaining the bound

Let

$$p = 2 - \sqrt{3}, \quad h = \sqrt{p}, \quad b = \sqrt{1-p}, \quad w = \frac{1-p}{\sqrt{1-2p}}, \quad z = \frac{h\sqrt{1-p}}{\sqrt{1-2p}}.$$

Define the real matrices

$$T_1 = \frac{1}{2} \begin{pmatrix} w+2h & b+z \\ -b-z & -w \end{pmatrix}, \quad T_2 = \frac{1}{2} \begin{pmatrix} 2h-w & b-z \\ -b+z & w \end{pmatrix}.$$

The identities $b^2 + z^2 = w^2$ and $bz = wh$ show that both determinants vanish. Directly, their nonzero singular values are $w + h$ and $w - h$, respectively. Applying (2) gives

$$\mathcal{S}(T_1) = \frac{w+h}{2}I + \frac{h}{2}\text{diag}(1, -1), \quad \mathcal{S}(T_2) = \frac{w-h}{2}I - \frac{h}{2}\text{diag}(1, -1).$$

Their sum is wI , whereas

$$T_1 + T_2 = \begin{pmatrix} 2h & b \\ -b & 0 \end{pmatrix}, \quad \mathcal{S}(T_1 + T_2) = \text{diag}(1+p, 1-p).$$

Therefore

$$\frac{\|\mathcal{S}(T_1 + T_2)\|_\infty}{\|\mathcal{S}(T_1) + \mathcal{S}(T_2)\|_\infty} = \frac{1+p}{w} = \sqrt{6\sqrt{3} - 9}.$$

Padding with zero summands handles every $m \geq 2$. This proves sharpness and completes the proof of Theorem 1.1. $\square$

References

- [1] Teng Zhang. Operator symmetric moduli and sharp triangle inequalities. arXiv:2603.01046, 2026. Version consulted: v1; Conjecture 1.6, equation (1.4), Problems 1.11 and 1.17. URL: <https://arxiv.org/abs/2603.01046>.